05.03.2026

SE 216 – SOFTWARE PROJECT MANAGEMENT
Spring 2025-2026

Project Proposal

NeuroMatter: Universal Local AI Hub

Problem Definition

Most smart home platforms fail to address the specific sensory needs of individuals with conditions such as ADHD, Dementia, or Autism. This project addresses the following issue: "How can we develop a closed-box, localized AI system that manages stressors for these populations while ensuring data sovereignty?

Background Information

Environmental triggers like excess light, noise, and vibrations cause severe distress for neurodivergent populations. Current cloud-based systems present significant privacy risks and frequently fail during internet outages, leaving vulnerable users without reliable support. Therefore, a localized, autonomous alternative is essential to provide a secure environment that processes sensory data entirely within the home network. This approach ensures that sensitive behavioral data remains private while providing continuous assistance for residents during their daily routines.

Objectives

- **Localized Privacy Sovereignty:** All processing occurs on a closed-box hub to eliminate cloud risks. This architecture ensures total data privacy within the home network.

- **Matter-Based Universal Integration:** We use the Matter protocol for brand-agnostic, local device communication. This ensures universal hardware compatibility without requiring an active internet connection.

- **AI-Powered Sensory Analysis:** Local AI models use image and sound processing to measure stressors. The system automatically adjusts environments to prevent sensory overload.

- **Offline Operational Reliability:** The localized software maintains all monitoring functions during internet outages.This removes dependency on external server stability and ensures reliable operation.

- **Autonomous Hazard Prevention:** The system identifies routine anomalies to issue immediate alerts for residents. This proactive approach prevents household accidents and ensures a secure environment.

Approval Signatures and GitHub Accounts

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